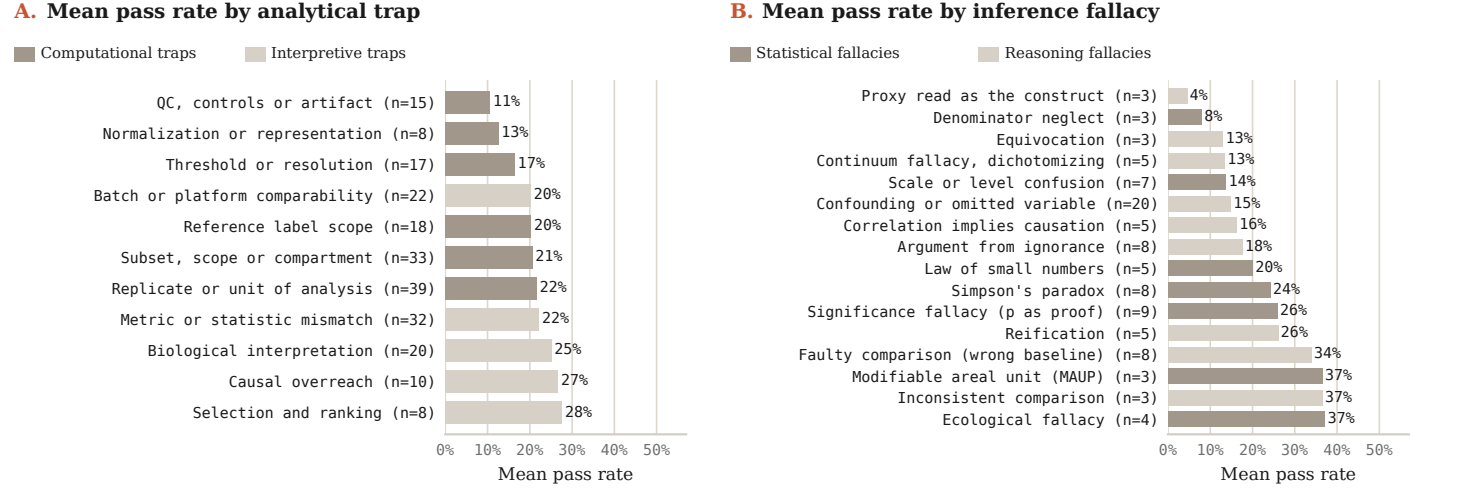
A. Mean pass rate by analytical trap
Computational traps
Interpretive traps
QC, controls or artifact (n=15)
11%
Normalization or representation (n=8)
13%
Threshold or resolution (n=17)
17%
Batch or platform comparability (n=22)
20%
Reference label scope (n=18)
20%
Subset, scope or compartment (n=33)
21%
Replicate or unit of analysis (n=39)
22%
Metric or statistic mismatch (n=32)
22%
Biological interpretation (n=20)
25%
Causal overreach (n=10)
27%
Selection and ranking (n=8)
28%
0%
10%
20%
30%
40%
50%
Mean pass rate
B. Mean pass rate by inference fallacy
Statistical fallacies
Reasoning fallacies
Proxy read as the construct (n=3)
4%
Denominator neglect (n=3)
8%
Equivocation (n=3)
13%
Continuum fallacy, dichotomizing (n=5)
13%
Scale or level confusion (n=7)
14%
Confounding or omitted variable (n=20)
15%
Correlation implies causation (n=5)
16%
Argument from ignorance (n=8)
18%
Law of small numbers (n=5)
20%
Simpson's paradox (n=8)
24%
Significance fallacy (p as proof) (n=9)
26%
Reification (n=5)
26%
Faulty comparison (wrong baseline) (n=8)
34%
Modifiable areal unit (MAUP) (n=3)
37%
Inconsistent comparison (n=3)
37%
Ecological fallacy (n=4)
37%
0%
10%
20%
30%
40%
50%
Mean pass rate